

- 4 An object is held at rest just below the surface of a liquid of uniform density as shown in Figure 4.1.

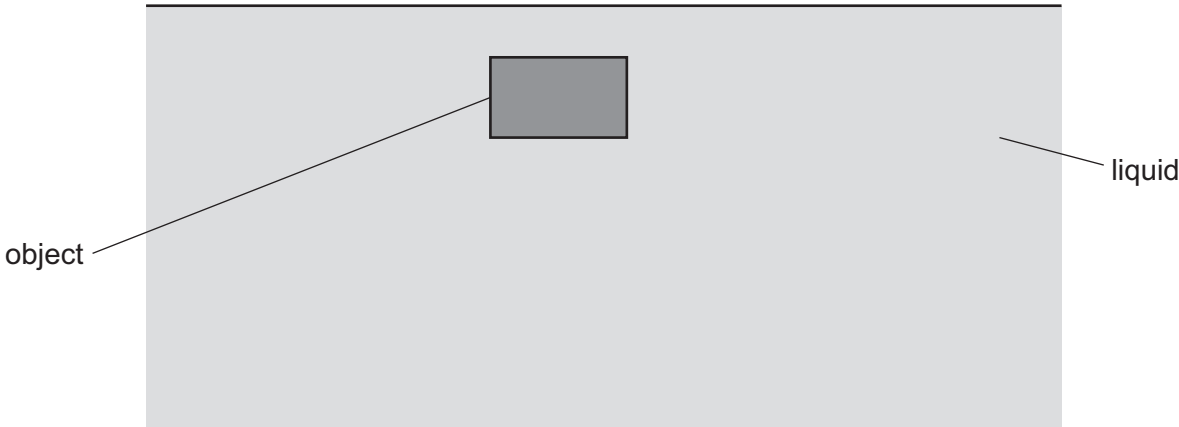


Figure 4.1

The object is released and accelerates downwards until it reaches terminal velocity.

The magnitude of the weight of the object is W .

The magnitude of the upthrust acting on the object is U .

The magnitude of the drag force acting on the object is D .

- (a) By placing **one** tick (✓) in each row, complete Table 4.1 to show what happens to the magnitude of the three forces acting on the object as it accelerates.

Table 4.1

force	increases	stays the same	decreases
W			
D			
U			

[3]

- (b) State an expression, in terms of W , D and U , that shows the relationship between the forces acting on the object when the object is at its terminal velocity.

[1]

(c) The object has a mass of 0.070 kg and a volume of $1.0 \times 10^{-5} \text{ m}^3$.

The density of the liquid is 1100 kg m^{-3} .

Calculate D when the object is at terminal velocity.

$D = \dots\dots\dots \text{ N [3]}$

[Total: 7]